



High Performance Countercurrent Membrane Purification for protein separations

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ABSTRACT

Although pressure-driven membrane filtration is used extensively in bioprocessing, protein purification is currently performed by chromatographic separations due to the difficulty in achieving high selectivity for similarly sized species using commercially available ultrafiltration membranes in the absence of fouling. This work examines a novel approach for protein separations based on High Performance Countercurrent Membrane Purification (HPCMP), which exploits highly selective diffusive transport across the thin walls of a hollow fiber membrane. Experiments were performed using commercially available hemodialysis membranes using both polydisperse dextrans and a model protein mixture of bovine serum albumin (BSA) and myoglobin (Mb). Dextran transport data obtained using diffusive transport showed significantly greater selectivity than that for pressure-driven filtration. The HPCMP system was then used to separate BSA and Mb, achieving greater than 98% yield of both proteins with purification factors greater than 100-fold. The Mb product obtained in the draw (dialysate) solution had undetectable levels of BSA. Stable operation was achieved for 96 h without any evidence of membrane fouling due to the absence of any significant pressure-driven flow. These results demonstrate the feasibility of using HPCMP for high resolution separations in the preparation of biopharmaceuticals and natural protein products.

List of symbols

A	Effective Membrane Area (m^2)
a_n	Expansion coefficients for hindered convection
b_n	Expansion coefficients for hindered diffusion
C_B	Local concentration of solute in the bulk solution (g L^{-1})
C_D	Concentration of solute in the draw/dialysate (g L^{-1})
C_F	Concentration of solute in the feed/bulk (g L^{-1})
C_P	Concentration of solute in the permeate (g L^{-1})
C_R	Concentration of solute in the retentate (g L^{-1})
C_w	Concentration of solute at the membrane wall (g L^{-1})
D	Diffusion Coefficient ($\text{m}^2 \text{s}^{-1}$)
k_B	Boltzmann's Constant ($\text{m}^2 \text{kg s}^{-2} \text{K}^{-1}$)
K_c	Convective hindrance factor
K_d	Diffusive hindrance factor
k_m	Mass transfer coefficient (m s^{-1})
MW	Molecular weight of dextran or protein (Da)

$MWCO$	Molecular weight cutoff of membrane (Da)
N_s	Solution diffusion rate across the membrane ($\text{g m}^{-2} \text{s}^{-1}$)
P	Purification factor for species
q_F	Feed flow rate (mL min^{-1})
q_D	Dialysate (draw) flow rate (mL min^{-1})
q_P	Permeate flow rate (mL min^{-1})
R_f	Fractional removal of solute
r_p	Radius of pore (m)
r_s	Radius of solute (m)
S_o	Observed membrane sieving coefficient
T	Temperature (K)
Y	Yield

Greek Symbols

α	Ratio of dialysate to feed flow rates
δ_m	Membrane thickness (m)
ϵ	Membrane porosity
λ	Ratio of the solute radius to pore radius

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μ	Solution viscosity (Pa s)
φ	Equilibrium partition coefficient into pore

1. Introduction

Tangential Flow Filtration (TFF) is used in the commercial scale manufacture of essentially all biotherapeutics [1–6]. This includes the use of tangential flow microfiltration for initial clarification of cell culture fluid as well as ultrafiltration for final concentration and formulation [1,7,8]. However, both of these TFF processes involve systems in which there are relatively large differences in size, e.g., between cells/cell debris (typically 0.1–10 μm in size) and proteins (≈ 10 nm) or between proteins and buffer components (≤ 1 nm).

There has been considerable interest in the development of TFF processes for much higher resolution separations to replace expensive chromatographic processes in downstream bioprocessing; such processes are typically known as High Performance Tangential Flow Filtration or HPTFF [5,9–13]. As originally described, HPTFF provides highly selective separations by operating the pressure-driven TFF device in the pressure-dependent regime, i.e., at or below the “transition point” in a plot of filtrate flux as a function of transmembrane pressure [11]. Subsequent studies using a variety of model systems demonstrated that significant improvements in selectivity can be achieved by controlling pH and ionic strength to exploit differences in charge and effective volume between the product and impurities and by using electrically-charged membranes to enhance the repulsion of like-charged species [13–16]. High yield and purification factors were achieved by batch diafiltration to effectively wash impurities through the membrane while recirculating the highly retained product (or impurity). HPTFF has been successfully demonstrated in small-scale laboratory systems for the purification of an antigen binding fragment from *E. coli* host cell proteins [12], a monoclonal antibody from mammalian cell host proteins [17], and a pegylated product from residual protein and PEG after the conjugation reaction [18], among others.

Despite these efforts, there has been relatively little progress in the application of HPTFF in large scale commercial manufacturing of biopharmaceuticals. Although the reasons for this are varied, they include the lack of commercially available charged ultrafiltration membranes, challenges with membrane fouling associated with the pressure-driven filtration, the difficulty in obtaining sufficient selectivity between similarly-sized products and impurities, and the large buffer consumption required for the diafiltration process. In addition, HPTFF is performed as a batch process using fairly expensive membrane modules while the industry as a whole has been migrating towards single-use equipment and continuous purification processes [1,7,19,20]. This has created new opportunities for downstream processing given the inherent batch nature of bind-and-elute chromatography.

The objective of this study was to examine the potential of performing high resolution continuous protein separations using commercially available hollow fiber (HF) membranes operated with countercurrent contacting and diffusive transport across the membrane, a new approach that will be referred to as High Performance Countercurrent Membrane Purification or HPCMP. Data for this study were obtained using mixtures of bovine serum albumin (BSA) and myoglobin (Mb) as a model protein system; this protein mixture was used previously by Kumar and Ulbricht [21] and Nakatsuka and Michaels [22] in their studies of pressure-driven HPTFF. In addition, experiments performed with polydisperse dextrans were used to demonstrate the inherent increase in selectivity associated with the diffusive transport in the HPCMP configuration. The results obtained in this study clearly demonstrate the potential of using low-cost single-use hollow fiber membranes for continuous HPCMP for the purification of high value biopharmaceuticals and natural protein products.

2. Materials and methods

Experiments were performed with BSA (MW = 66 kDa) and Mb (MW = 17 kDa), both obtained from Sigma-Aldrich (Catalog numbers A2153 and MO630, respectively). Feed solutions were prepared by dissolving the BSA and Mb in 1x PBS buffer made by diluting a 10x concentrate (Thermo Scientific, Waltham, MA) with DI water to yield a final solution with conductivity of approximately 15 mS/cm and pH 7.8. Sodium azide was added at 0.02% w/v to prevent microbial growth during long experimental runs. All solutions were pre-filtered through 0.2 μm polyvinylidene fluoride (PVDF) membranes prior to use to remove undissolved protein aggregates. The high ionic strength of the PBS buffer (150 mM) likely reduces any electrostatic interactions between the negatively-charged BSA (isoelectric point of 4.7–5.0 [23,24], although some reports show a pI as high as 5.56 [25]). Mb is very weakly negatively charged at pH 7.8 (isoelectric point of 6.5–7.0 [26]). Future studies will examine the potential of exploiting protein-membrane electrostatic interactions to further enhance the protein separation as discussed by Castro-Muñoz et al. [27].

Protein separations were performed using Purema™ H hollow fiber dialyzers from 3 M Company (St. Paul, MN) containing 13,200 fibers made from a blend of polyethersulfone (PES) and polyvinylpyrrolidone (PVP). The fibers have an inner diameter of 200 μm , wall thickness of approximately 30 μm , and active length of 23.3 cm, giving a surface area of 1.9 m². Modules were operated in a vertical orientation with the dialysate solution introduced into the top port on the shell side of the module as shown in Fig. 1.

The protein feed was introduced into the lumen-side inlet at the bottom of the module, i.e., in countercurrent flow to the dialysate (draw) solution. Flow rates were controlled with Masterflex L/S peristaltic pumps (Cole-Parmer, Vernon Hills, IL) fitted with Tygon® E-LFL tubing (Cole-Parmer). A single pump equipped with two pump heads was used to control the inlet and outlet flowrates of the dialysate and thus maintain close to zero net ultrafiltration. Pumps were calibrated before each experiment, with the flow rates evaluated during the experiment by timed collection. Pressure gauges (Ashcroft, Stratford, CT) were placed immediately before/after the inlet and outlet ports. Before each experiment, the module was completely flushed with PBS buffer. The

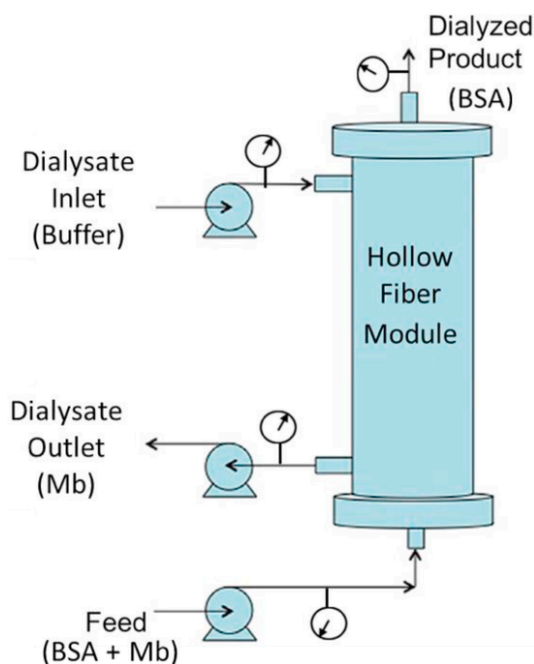


Fig. 1. Schematic of continuous countercurrent membrane purification system for protein separations.

feed was then replaced with the protein mixture with samples collected periodically from the feed, dialysate exit, and retentate exit lines for evaluation of the concentrations of the BSA and Mb concentrations.

BSA and Mb concentrations were evaluated by UV absorbance at 280 and 408 nm using an Infinite® m200 Pro microplate reader (Tecan Trading AG, Switzerland). Calibration curves were constructed for each protein by measuring the absorbance at both wavelengths for a series of samples at known concentrations. More detailed analysis of the protein mixtures was performed using an Agilent 1260 infinity II HPLC system (Agilent Scientific instruments, Santa Clara, CA) with Ultrahydrogel 2000, 500 and 120 size exclusion columns in series (Waters Corp, Milford, MA); a guard column was placed immediately before the Ultrahydrogel 2000 column. The HPLC was operated at a constant running buffer (PBS) flow rate of 0.6 mL/min, with the column temperature maintained at 30 °C. Samples were injected every 60 min and detected by a refractive index detector.

At the end of the experiment, the myoglobin in the draw solution was collected and concentrated by ultrafiltration using the same apparatus but without any dialysate [28]. Ultrafiltration was performed in a batch mode by recycling the retentate back into the feed with continuous removal of permeate. The system was operated at a feed flow rate of 300 mL/min with a total volume reduction of approximately 15-fold. The re-concentrated Mb was used in subsequent experiments to minimize protein cost.

Limited experiments were also performed using polydisperse dextrans in both HPCMP and pressure-driven (ultrafiltration) modes. Ultrafiltration was performed without any dialysate solution with the dialysate exit pump used to control the ultrafiltration rate. The dextran feed was prepared by mixing dextrans of various MWs including 0.2 g/L of Dextran T10 (9–11 kDa), 0.2 g/L of Dextran T40 (35–45 kDa), 0.8 g/L of Dextran T150 (150 kDa), and 1.8 g/L of Dextran T2000 (2000 kDa), obtained from Sigma-Aldrich (St. Louis, MO) and TCI Chemical (Portland, OR). Column calibration was performed using EasiVial polyethylene oxide (PEO) standards (Agilent Scientific instruments) as discussed by Yehl and Zydney [29].

3. Results and analysis

3.1. Membrane characterization

The transport properties of the hollow fiber membranes were first examined using polydisperse dextrans. Experiments were performed using both filtration and HPCMP modes of operation. For the filtration experiments, the data are plotted as the fractional transmission through the membrane, which is equal to the membrane sieving coefficient:

$$S = \frac{C_p}{C_f} \quad (1)$$

where C_p and C_f are the dextran concentrations in the permeate and feed solutions, respectively. The system was operated using $q_F = 500$ mL/min and $q_p = 200$ mL/min based on results presented by Yehl and Zydney [29]. The HPCMP experiments were performed using phosphate buffer as the draw solution using $q_F = 0.5$ mL/min and $q_D = 6.1$ mL/min, with the fractional dextran removal evaluated as:

$$R_f = 1 - \frac{C_R}{C_F} \quad (2)$$

where C_R is the dextran concentration in the retentate exit stream. In both cases, the concentrations of the different molecular weight dextrans were evaluated by HPLC based on the area under narrow slices of the chromatogram. Data were obtained after the solute concentrations in the permeate and retentate solutions had stabilized; this required about 15 min for the sieving (pressure-driven filtration) and 8-hr for

the HPCMP (due to the lower flow rates and the slower rate of dextran diffusion).

The dextran sieving coefficients and fractional removal rates were both equal to 100% for the very small dextrans which are easily transported across the membrane but decrease to zero for very large dextrans that are fully retained by the membrane pores. The HPCMP results are shifted to the right and appear much steeper than the sieving curve, indicating that diffusive transport has the potential to provide for a more selective separation than is possible using pressure-driven filtration. For example, if we look at conditions that provide sieving coefficients and fractional removal rates of 80 and 20%, the sieving curve yields a MW ratio for the corresponding dextrans of 2.5 while the HPCMP data yield a MW ratio of only 1.5.

3.2. Hydrodynamic analysis

In order to understand the greater selectivity for HPCMP compared to pressure-driven ultrafiltration, the sieving curve was directly evaluated using available hydrodynamic models for solute convection through porous membranes [25]:

$$S = \varphi K_c = \frac{\varphi (2 - \varphi) K_s}{K_t} \quad (3)$$

where φ is the equilibrium partition coefficient between the membrane pores and external solution immediately adjacent to the membrane:

$$\varphi = (1 - \lambda)^2 \quad (4)$$

and K_c is the hindrance factor for convection, which can be expressed in terms of the functions:

$$K_t = \frac{9}{4} \pi^2 \sqrt{2} (1 - \lambda)^{-\frac{5}{2}} \left[1 + \sum_{n=1}^2 a_n (1 - \lambda)^n \right] + \sum_{n=3}^7 a_n \lambda^n \quad (5a)$$

$$K_s = \frac{9}{4} \pi^2 \sqrt{2} (1 - \lambda)^{-\frac{5}{2}} \left[1 + \sum_{n=1}^2 b_n (1 - \lambda)^n \right] + \sum_{n=3}^7 b_n \lambda^n \quad (5b)$$

where the expansion coefficients a_n and b_n are provided by Bungay and Brenner [30] and summarized in Table A1. Both K_s and K_t are functions of λ , the ratio of the solute to pore radius:

$$\lambda = \frac{r_s}{r_p} \quad (6)$$

The solute radius can be expressed in terms of the solute diffusion coefficient using the Stokes-Einstein equation:

$$r_s = \frac{k_B T}{6\pi\mu D} \quad (7)$$

where k_B is Boltzmann's constant, T is the absolute temperature, μ is the solution viscosity, and D is the solute diffusion coefficient [31]. The dextran diffusion coefficient is directly related to the dextran molecular weight (MW) using literature correlations [31].

$$\log_{10}(D) = -8.1154 - 0.47752 \log_{10}(MW) \quad (8)$$

where D is in m^2/s and MW is in Da. The dashed curve in Fig. 2 is developed using $r_p = 3.7$ nm as determined by eye. The model calculation is somewhat steeper than the experimental data since the model assumes a perfectly uniform pore size in contrast to the pore size distribution that is present in the Purema H membrane.

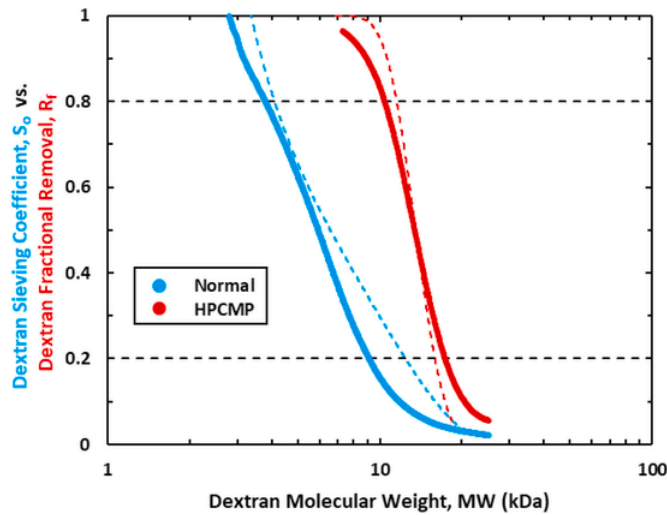


Fig. 2. Observed dextran sieving coefficients and fractional dextran removal as a function of dextran molecular weight for the hollow fiber module operated under pressure-driven flow (normal) and HPCMP conditions. The horizontal dashed lines represent arbitrarily targeted selectivity conditions to evaluate the MW ratio of the two curves. The dashed curves represent model calculations as described in the text.

The evaluation of the fractional dextran removal during HPCMP operation is a bit more complicated. The rate of solution diffusion across the hollow fiber membrane can be written as:

$$N_s = \frac{\varepsilon \phi K_d D}{\delta_m} (C_B - C_D) \quad (9)$$

where $\phi K_d D$ is the effective solute diffusion coefficient, ε is the membrane porosity, δ_m is the membrane thickness, and C_B and C_D are the solute concentrations in the bulk (feed) and draw solutions, respectively. The hindrance factor for diffusion (K_d) is given as [30]:

$$K_d = \frac{6\pi}{K_t} \quad (10)$$

with K_t given by Equation (5b).

The solute concentration profiles in the hollow fiber module can be evaluated by solving the appropriate mass balances for the bulk and draw solutions accounting for the countercurrent flow. The final equations are equivalent to those developed for countercurrent dialysis, with the fractional removal given as [32]:

$$R_f = 1 - \frac{C_R}{C_F} = \frac{1 - \exp(-\beta A)}{1 - \frac{\exp(-\beta A)}{\alpha}} \quad (11)$$

where A is the membrane area, α is the ratio of the dialysate to feed flow rates,

$$\alpha = \frac{q_D}{q_F} \quad (12)$$

and

$$\beta = \frac{\varepsilon \phi K_D D}{\delta_m q_F} \left(1 - \frac{1}{\alpha} \right) \quad (13)$$

The red dashed curve in Fig. 2 is the model calculation for the fractional removal using the same membrane pore radius along with the measured membrane area ($A = 1.9 \text{ m}^2$) and membrane thickness ($\delta_m = 30 \text{ }\mu\text{m}$) assuming a membrane porosity of $\varepsilon = 0.4$. The model results are in good agreement with the experimental data obtained using HPCMP operation (with $q_F = 0.5 \text{ mL/min}$ and $q_D = 6.1 \text{ mL/min}$

corresponding to $\alpha = 12.2$). The predicted values of the fractional removal rate are somewhat steeper than that experimental results, which is again due to neglecting the effects of the pore size distribution on solute transport. The much steeper slope for the fractional removal rate arises from the much stronger dependence of the hindrance factor for diffusion on the ratio of the solute to pore radii as given by the expressions for K_c and K_d (Equations 3 and 10, respectively).

3.3. BSA /Mb separation

The steep slope for the fractional removal rate seen in Fig. 2 suggests that it should be possible to use these hollow fiber membranes for the separation of proteins with relatively small differences in effective size. Experiments were performed using the model system of BSA and Mb since that protein combination was used in several previous studies of HPTFF [21,22]. Typical data obtained at a constant feed flow rate of $q_F = 2.0 \pm 0.1 \text{ mL/min}$ and $q_D = 8.0 \pm 0.1 \text{ mL/min}$, corresponding to $\alpha = 4.0$, with feed concentrations of $C_{BSA} = 40 \text{ g/L}$ and $C_{Mb} = 500 \text{ mg/L}$ are shown in Fig. 3. The mean transmembrane pressure difference under these conditions was less than 3 kPa throughout the experiment, corresponding to an ultrafiltration rate of less than 0.1 mL/min. The pressure drop across the module (feed inlet minus retentate exit) was also less than 3 kPa due to the low feed flow rate (2.0 mL/min) used in this experiment; typical hemodialysis is performed using a blood-side flow rate around 300 mL/min. There was no evidence of any membrane fouling over the 9-hr experiment, with ex-

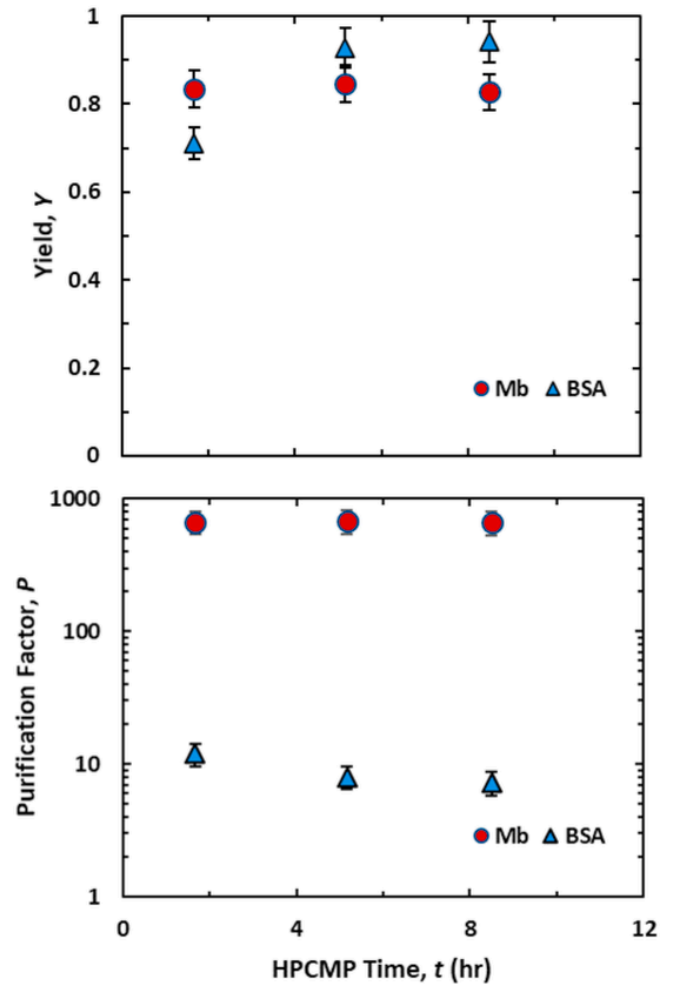


Fig. 3. Yield and Purification factor of Mb and BSA. Data were obtained at a Mb and BSA feed concentrations of 500 mg/L and 40 g/L and at feed and dialysate flow rates of 2.0 ± 0.1 and $8.0 \pm 0.2 \text{ mL/min}$, respectively.

cellent mass balance closure (within 5%) and nearly constant concentration and pressure profiles. This is consistent with previous studies using these hollow fiber membranes for buffer exchange [33] and concentration [28], with the absence of any fouling due to the very low pressure-driven filtration flow in all of these studies.

The results are plotted as the protein yield:

$$Y_{BSA} = \frac{q_R C_{BSA,R}}{q_F C_{BSA,F}} Y_{Mb} = \frac{q_D C_{Mb,D}}{q_F C_{Mb,F}} \quad (14)$$

and purification factor:

$$P_{BSA} = \frac{\left(\frac{C_{BSA,R}}{C_{Mb,R}}\right)}{\left(\frac{C_{BSA,F}}{C_{Mb,F}}\right)} \quad P_{Mb} = \frac{\left(\frac{C_{Mb,D}}{C_{BSA,D}}\right)}{\left(\frac{C_{Mb,F}}{C_{BSA,F}}\right)} \quad (15)$$

with the BSA and Mb concentrations determined by UV absorbance. The Mb yield in the dialysate was relatively constant at 82%. The BSA yield based on the A280 measurements for the retentate exit stream increased over the first few hours, stabilizing at a value > 95% at longer times. However, samples of the dialysate showed no detectable BSA throughout the HPCMP process. A separate experiment was thus performed with a high concentration of only BSA in the feed solution using $\alpha = 1$, with concentrations evaluated using HPLC. The results showed a very small amount of BSA in the dialysate consistent with a fractional removal of approximately 0.3%. Based on this result and the value of α used in the BSA-Mb separation experiment, the BSA yield is expected to be >99%, which is consistent with the HPLC chromatogram presented in Fig. 4. The purification factor for BSA for this experiment was ≥ 7 -fold, but the estimated Mb purification factor was >600-fold based on the data obtained with only BSA in the feed.

In order to increase the removal of Mb from the retentate, a second HPCMP separation was performed but with a larger value of $\alpha = 12.2$ based on $q_F = 0.5 \pm 0.1$ mL/min and $q_D = 6.1 \pm 0.3$ mL/min. The pressure drop across the module (feed inlet minus retentate exit) was less than 3 kPa under these conditions due to the very low feed flow rate. The Mb in the dialysate had a nearly constant purification factor of 300-fold throughout the experiment with $Y_{Mb} = 98\%$, which was significantly higher than the 82% seen in Fig. 3 due to the lower q_F and higher value of α . The low yield and purification factor in the first 4 h is largely due to dilution effects associated with washing out the hold-up volume within the module, which was initially filled with protein-free buffer. It is also possible that there is some low level of fouling at the start of the experiment, although there was no independent evidence for this phenomenon. The greater Mb removal provided a steady-state purification factor for BSA of approximately 70-fold, a factor of 10x greater than obtained in Fig. 3.

In order to confirm the high degree of purification seen in Fig. 4, samples were collected from the feed, dialysate, and retentate exit lines towards the end of each experiment and analyzed by HPLC with results shown in Fig. 5. The feed shows two distinct peaks centered at 41 and 43 min, corresponding to BSA and Mb, respectively. The retentate samples shows a large BSA peak with a “tail” at longer times which is likely associated with the presence of a small amount of Mb. The dialysate solution shows only a single peak corresponding to Mb with no evidence of any BSA under these conditions.

One of the potential advantages of HPCMP is that the predicted performance should be relatively independent of the feed concentrations for either the product or impurity. This is in sharp contrast to pressure-driven TFF processes in which high concentrations lead to low filtrate flux due to the build-up of a highly concentrated region of retained solutes at the membrane surface. In order to confirm this result, a subsequent experiment was performed at much higher protein concentrations: $C_{Mb} = 1140$ mg/L and $C_{BSA} = 50$ g/L. Data were obtained over a 96-hr run with the same module (without any cleaning or regeneration), beginning at low concentrations (Experiment 3 in Table 1) and

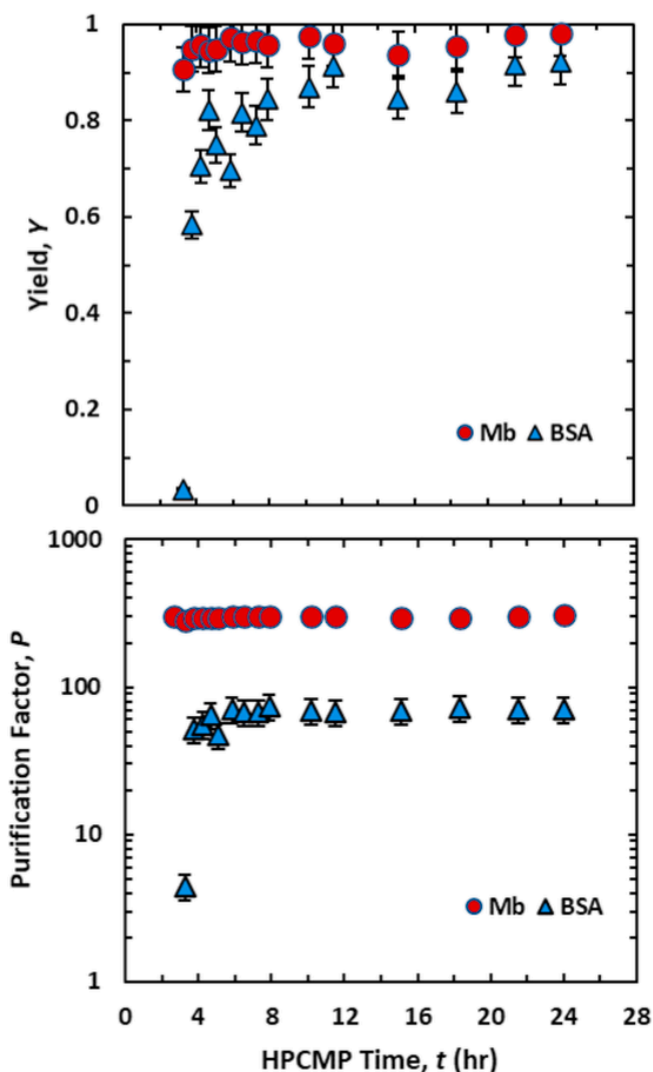


Fig. 4. Yield and Purification factor of Mb and BSA. Data were obtained at a Mb and BSA feed concentrations of 450 mg/L and 2 g/L and at feed and dialysate flow rates of 0.5 ± 0.1 and 6.1 ± 0.3 mL/min, respectively.

then moving to the higher concentrations after 48-hr (Experiment 4). Results for the steady-state yield and purification factor for these, and the other HPCMP experiments performed in this study, are summarized in Table 1. None of the experiments showed any detectable BSA in the dialysate based on the absorbance measurements, with the purification factor for Mb being > 300-fold under all conditions. The data for the 3 experiments with $\alpha \approx 12$ showed greater than 98% yield of Mb in the dialysate. The results from Experiment 4, using the 50 g/L BSA, showed a slightly lower Mb yield than that obtained in Experiment 3, although these values were comparable to the Mb yield in Experiment 2. This confirms the effectiveness of the HPCMP process even at high protein concentrations. It is also worth noting that multiple experiments were performed with the same hollow fiber module, with the module cleaned between experiments simply by flushing with DI water and then buffer. There was no evidence of any decrease in performance for the previously used modules, providing further support for the absence of any significant fouling in the HPCMP experiments.

4. Discussion

The data presented in this study provide the first demonstration of High Performance Countercurrent Membrane Purification (HPCMP) for the separation of a model protein system. Data obtained with BSA and

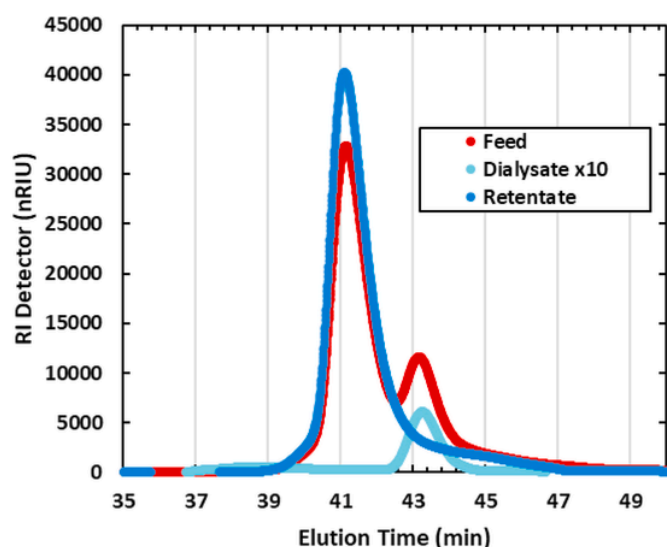


Fig. 5. HPLC analysis of feed, dialysate, and retentate samples at the end of the BSA-Mb separation. The RI signal for the dialysate was multiplied by 10x due to the dilution effects with $\alpha = 12.2$.

Table 1
Summary of HPCMP results for BSA-Mb separation.

Experiment	C_{BSA} (g/L)	C_{Mb} (g/L)	q_F (mL/min)	q_D (mL/min)	α	Y_{Mb}	P_{Mb}
1	40	0.5	2.0	8.0	4.0	82%	> 600
2	2	0.45	0.5	6.1	12	98%	> 300
3	2	0.5	0.5	6.4	13	99.5%	> 300
4	50	1.1	0.5	6.4	13	98%	> 300

Mb showed >98% yield of both proteins with purification factors for Mb > 300-fold while that for BSA was as high as 70-fold. Stable performance was maintained for 96-hr of continuous separation, with no evidence of any fouling over the course of the experiments. Note that previous studies of BSA – Mb separation using pressure-driven membrane separations reported a maximum selectivity of only 14.2 [22], and this still required the use of specially-designed negatively-charged membranes made with sulfonic-acid modified carbon nanotubes. Data obtained with conventional ultrafiltration membranes (without the carbon nanotubes) showed a maximum selectivity of less than 7-fold [22], which is more than 40 times smaller than the selectivity obtained in the HPCMP experiments performed in this study.

The high selectivity seen in HPCMP was confirmed using dextran transport measurements obtained with the same membrane module but operated using either pressure-driven filtration or countercurrent diffusion. Results for the HPCMP system showed a significantly greater selectivity than that for pressure-driven filtration, consistent with theoretical calculations performed using available hydrodynamic models for solute convection and diffusion across porous membranes.

Although the data obtained in this study were for small laboratory-scale separations, the scale-up of this work should be relatively straightforward. The HPCMP process described in Experiment 1 used a BSA mass flow rate of 80 mg/min, corresponding to 0.12 kg/day. This is approximately the same output as provided by a 100 L perfusion bioreactor operated at a perfusion rate of 1 reactor volume/day with a product titer of 1.2 g/L. Scale-up to a 500 L single-use bioreactor would simply require 5 hollow fiber modules in parallel, with a total cost of $\leq \$100$. In addition, the buffer requirement for this experiment

(with $\alpha = 4$) is significantly less than that required for batch HPTFF processes.

Another unique advantage of HPCMP is that the system operates continuously with only a single pump pass. In contrast, previous studies of pressure-driven HPTFF were all performed using batch diafiltration, with the retained product recirculated through the membrane module (and the feed pump) more than 100 times during the process. This repeated pumping can reduce product quality, e.g., through the formation of protein aggregates, an effect that should be negligible in HPCMP. The ability to operate HPCMP continuously for extended periods of time provides exciting opportunities for use of this technology as part of a fully-integrated continuous process for biopharmaceutical manufacturing with lower capital and operating costs, increased manufacturing flexibility, and greater speed to market [34–38]. Future studies will be required to more fully explore the opportunities for optimizing the performance, identifying the most attractive applications, and effectively integrating HPCMP into the downstream purification process for high-value biopharmaceuticals and natural products.

5. Conclusions

High Performance Countercurrent Membrane Purification (HPCMP) has been shown to provide highly selective protein separations based on diffusive protein transport across conventional hollow fiber hemodialysis membranes. Mixtures of bovine serum albumin and myoglobin were separated with purification factors of greater than 100-fold and yields greater than 98% over 96 h of continuous operation. HPCMP has the potential to provide an exciting new technology for biomanufacturing, enabling continuous (steady-state) purification in an easily scalable and low-cost platform for downstream bioprocessing.

CRedit authorship contribution statement

Christopher J. Yehl: Data curation, Formal analysis, Investigation, Writing – original draft. **Andrew L. Zydney:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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